Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/02/2025
PI Name:	Sundaresan, Alamelu Ph.D.		
Project Title:	Antibody Somatic Hypermutation Under Spaceflight Conditions		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	77004	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 ILSRA--Flight Opportunities for Space Life Sciences (non-US proposers)
Start Date:	09/01/2020	**End Date:**	08/31/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 08/31/2026 per PI and NSSC information (Ed., 9/8/25) NOTE: End date changed to 08/31/2025 per F. Hernandez/ARC (Ed., 8/29/24) NOTE: End date changed to 08/31/2024 per PI and NSSC information (Ed., 9/2/23)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Frippiat, Jean-Pol Ph.D. (European Space Agency (ESA) PI: Lorraine University, France)		
Grant/Contract No.:	80NSSC20K1308		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Dr Alamelu Sundaresan is the U.S. Co-Investigator on this European Space Agency (ESA)-sponsored project. ESA/CNES (Centre National d'Etudes Spatiales/ French National Centre for Space Studies) Principal Investigator is Jean-Pol Frippiat, Ph.D., EA7300, Lorraine-University, France. Immunity is a key biological system affected by spaceflight. Our previous space experiment suggested that spaceflight conditions decrease the frequency of antibody somatic hypermutations (Bascove et al., 2011, FASEB Journal). This observation could imply less efficient maturation of antibody affinity in space, which would in turn decrease the efficiency of the immune response. Because these data were obtained using a small number of amphibians (n=2), we would like to 1) confirm these results using a constitutively hypermutating human B cell line and 2) understand how the genetic machinery that introduces these mutations is altered in space. To reach these goals, we plan to culture a constitutively hypermutating human B cell line in the International Space Station (ISS). Cells will be grown onboard the ISS and samples of this culture will be fixed in-flight to determine spaceflight-induced changes. The same experiment will be performed on Earth as a control. Space samples will be returned, when possible, to laboratories where they and control samples will be used for 1) somatic hypermutation analyses to determine if their frequency and pattern along an immunoglobulin VDJ (variability, diversity, and joining) rearrangement are modified and 2) gene expression profiling using microarrays to understand how the somatic hypermutation machinery that introduces these mutations is affected by spaceflight conditions. These analyses will expand our knowledge regarding the effects of spaceflight conditions on antibody somatic hypermutations and consequently, contribute to a better understanding of spaceflight-associated weakening of the immune system, which is an important issue for the preparation of long-term crewed space missions. Furthermore, because several studies have suggested that age-related deficiencies in the generation of high-affinity antibodies might be due to impaired somatic hypermutation, this project could also provide useful data for understanding age-related changes in the immune system. The ISS flight experiment will be manifested in future and the experiment time line is thus tentative and could be altered. Bascove M, Guéguinou N, Schaerlinger B, Gauquelin-Koch G, Frippiat JP. Decrease in antibody somatic hypermutation frequency under extreme, extended spaceflight conditions. FASEB J. 2011 Sep;25(9):2947-55. https:// . PubMed PMID: 21593434.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These analyses will improve our knowledge about the effects of spaceflight conditions on antibody somatic hypermutations and consequently contribute to a better understanding of spaceflight-associated immune system weakening, an important issue for the preparation of long-term manned space missions. Furthermore, since several studies suggested that age-related deficiency in generating high-affinity antibodies might be due to impaired somatic hypermutations, this project could also provide useful data to understand age-related changes of the immune system.
Task Progress:	We have found signaling pathways perturbed related to B cells in our Luminex multianalyte profiling technique, which we are perfecting in the lab. Protein expression of RANTES, TNF alpha, and VEGF alpha are upregulated to a higher degree in lunar gravity and modeled microgravity. This is a novel finding in these cells. We are investigating a new protein marker for somatic hypermutation. This was not known earlier and will be a good addition/alternative to the sequencing protocol followed to analyze hypermutation at present. We will also work on other molecules downstream of this marker in the B cells.
Bibliography Type:	Description: (Last Updated: 07/02/2026)
Books/Book Chapters	Mann V, Sundaresan A. "Regenerative medicine and spaceflight." in "Precision Medicine for Long and Safe Permanence of Humans in Space." Ed. C. Krittanawong. Academic Press, 2025. p.163-77. https://doi.org/10.1016/B978-0-443-22259-7.00039-4 , Jan-2025